TidyTuesday Week 42

Pranay Gundam

Tuesday 13th October, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the only recurring feature in economics where we throw two completely unrelated time series into a blender and solemnly report whatever sludge comes out. This week the dice gods handed us a short, two-day docket, which means I had twice the free time to overreact to things that mean nothing. Let us proceed with the appropriate gravity.

Monday, October 12, we paired the DISCONTINUED U.S. Balance on Services with the exchange rate of the Belizean dollar against the U.S. dollar, because of course we did. The raw regression came back "significant" with a p-value of 0.045, squeaking under the sacred 0.05 threshold like a student submitting at 11:59 PM, and explained a thrilling nine percent of the variance. Normally this is where I'd warn you that the detrended version would vaporize it into dust. But plot twist: the detrended relationship actually got a little stronger (p around 0.018, R-squared up to twelve percent), and the sign flipped from positive to negative, which is the regression equivalent of a character coming back from the dead with a completely different personality. So, technically, this one survives detrending, which makes it one of the project's rarer "huh, that held up" moments. Do I believe that quarterly U.S. service trade and the Belizean exchange rate are secretly intertwined? No. Twelve percent of anything is not a love story. But it's more than I usually get, so I'm framing it.

Tuesday, October 13, delivered the actual star of the week: Southern gasoline prices in the late '70s-to-'90s versus the 30-year FHA mortgage rate. The raw regression was strong, explaining about seventeen percent of the variance with a p-value you'd need a microscope to read. And then the detrended version did something genuinely impressive — it didn't collapse, it ascended. R-squared rocketed to a frankly indecent 0.63, with a p-value that has more zeros after the decimal than my patience has limits. For this project, a detrended R-squared over sixty percent is practically a UFO sighting. Now, before anyone drafts a Nobel acceptance speech: these are both things that swing around with 1970s-and-'80s inflation and interest-rate chaos, so a shared macroeconomic mood is a far more boring and likely explanation than gasoline whispering sweet nothings to mortgage underwriters. Still, as random pairings go, this is the closest thing to "a real relationship" we've stumbled into in a while.

As always, the obligatory disclaimer that I am legally and morally required to shout: these series are paired at random, nobody is claiming anything causes anything, and spurious correlation is not the bug here — it is the entire genre. The whole point of this exercise is to bask in the statistical nonsense that falls out when you let a computer matchmake economic data with its eyes closed. So enjoy the gasoline-mortgage romance for what it is: a pretty number, a shared

historical vibe, and absolutely no reason to restructure your portfolio. See you next week, when the odds strongly favor a return to our regularly scheduled garbage.

2 Date: 2026-10-12

Series ID: BOPBSVN
This series is titled Balance on Services (DISCONTINUED) and has a frequency of Quarterly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1960-01-01 and the observation end date is 2014-01-01.The popularity of this series is 1.

Series ID: XRNCUSBZA618NRUG
This series is titled Exchange Rate (market+estimated) for Belize and has a frequency of Annual. The units are National Currency Units per US Dollar and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1970-01-01 and the observation end date is 2023-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_XRNCUSBZA618NRUG	**R-squared:**	0.090
Model:	OLS	**Adj. R-squared:**	0.069
Method:	Least Squares	**F-statistic:**	4.246
Date:	Mon, 12 Oct 2026	**Prob (F-statistic):**	0.0454
Time:	20:28:58	**Log-Likelihood:**	34.026
No. Observations:	45	**AIC:**	-64.05
Df Residuals:	43	**BIC:**	-60.44
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.9274	0.024	79.937	0.000	1.879	1.976
value_fred_BOPBSVN	0.0023	0.001	2.060	0.045	4.83e-05	0.004

Omnibus:	17.591	**Durbin-Watson:**	0.425
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	23.370
Skew:	-1.275	**Prob(JB):**	8.42e-06
Kurtosis:	5.441	**Cond. No.**	30.5

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

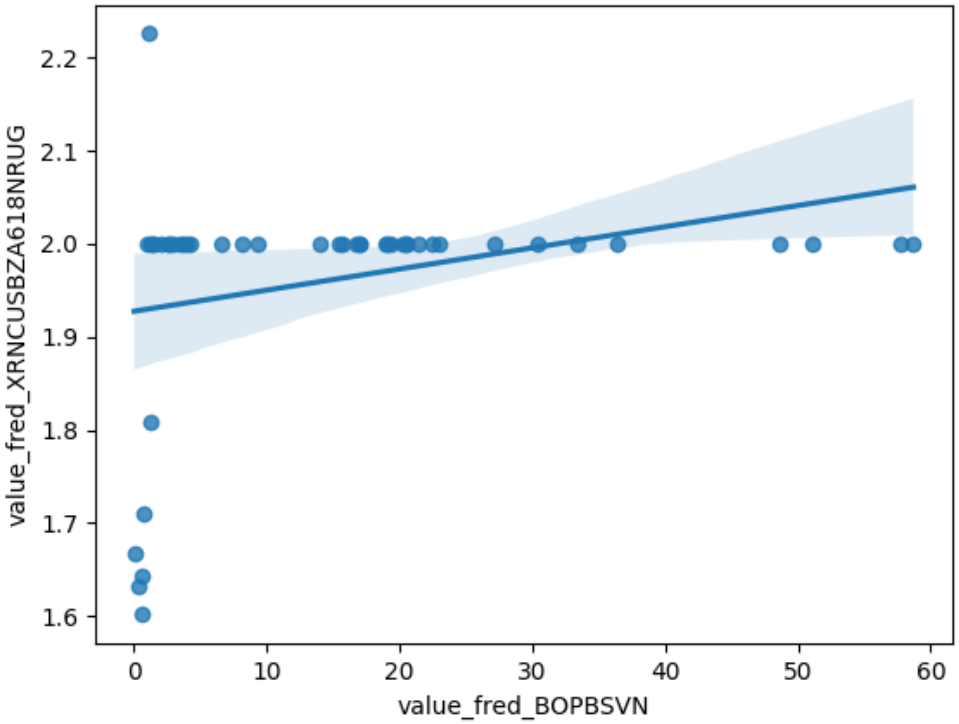


Figure 1: Raw Regression Plot for 2026-10-12

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_XRNCUSBZA618NRUG_detrended	**R-squared:**	0.123
Model:	OLS	**Adj. R-squared:**	0.103
Method:	Least Squares	**F-statistic:**	6.041
Date:	Mon, 12 Oct 2026	**Prob (F-statistic):**	0.0181
Time:	20:28:58	**Log-Likelihood:**	41.263
No. Observations:	45	**AIC:**	-78.53
Df Residuals:	43	**BIC:**	-74.91
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	5.671e-15	0.015	3.84e-13	1.000	-0.030	0.030
value_fred_BOPBSVN_detrended	-0.0050	0.002	-2.458	0.018	-0.009	-0.001

Omnibus:	8.915	**Durbin-Watson:**	0.588
Prob(Omnibus):	0.012	**Jarque-Bera (JB):**	16.952
Skew:	0.296	**Prob(JB):**	0.000208
Kurtosis:	5.948	**Cond. No.**	7.28

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

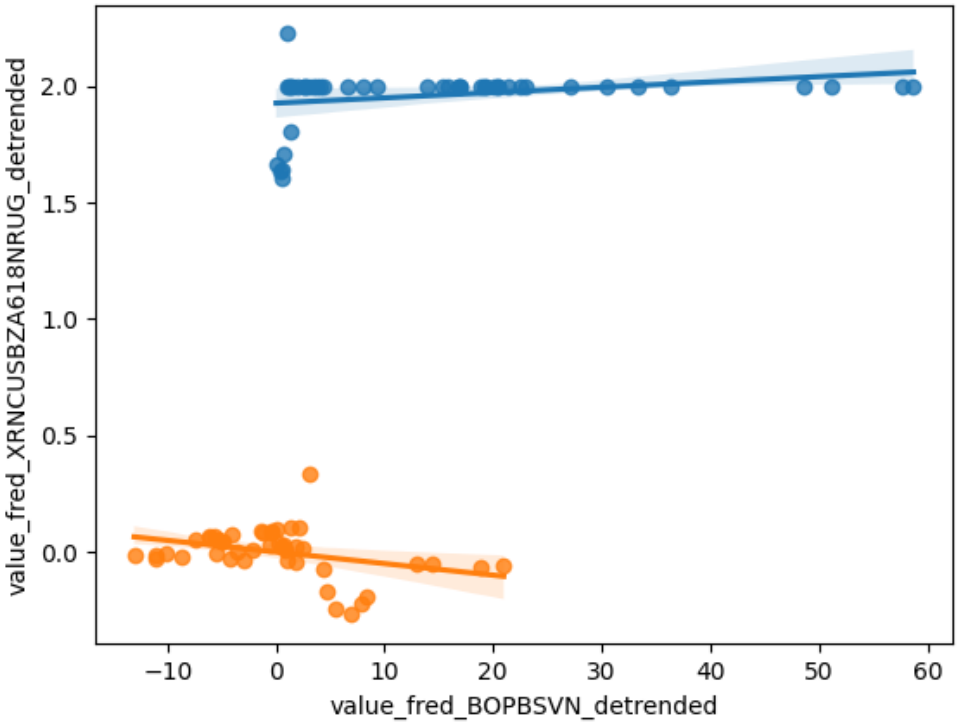


Figure 2: Detrended Regression Plot for 2026-10-12

3 Date: 2026-10-13

Series ID: APUB30074714
This series is titled Average Price: Gasoline, Unleaded Regular (Cost per Gallon/3.785 Liters) in the South Census Region - Size Class B and has a frequency of Monthly. The units are U.S. Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1978-01-01 and the observation end date is 1997-12-01.The popularity of this series is 1.

Series ID: FHA30
This series is titled 30-Year FHA Mortgage Rate: Secondary Market (DISCONTINUED) and has a frequency of Monthly. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1964-01-01 and the observation end date is 2000-06-01.The popularity of this series is 9.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_FHA30	**R-squared:**	0.169
Model:	OLS	**Adj. R-squared:**	0.165
Method:	Least Squares	**F-statistic:**	47.29
Date:	Tue, 13 Oct 2026	**Prob (F-statistic):**	5.59e-11
Time:	18:35:32	**Log-Likelihood:**	-534.79
No. Observations:	235	**AIC:**	1074.
Df Residuals:	233	**BIC:**	1081.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	3.5713	1.048	3.408	0.001	1.507	5.636
value_fred_APUB30074714	6.5135	0.947	6.876	0.000	4.647	8.380

Omnibus:	11.293	**Durbin-Watson:**	0.039
Prob(Omnibus):	0.004	**Jarque-Bera (JB):**	6.092
Skew:	0.195	**Prob(JB):**	0.0476
Kurtosis:	2.315	**Cond. No.**	13.6

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

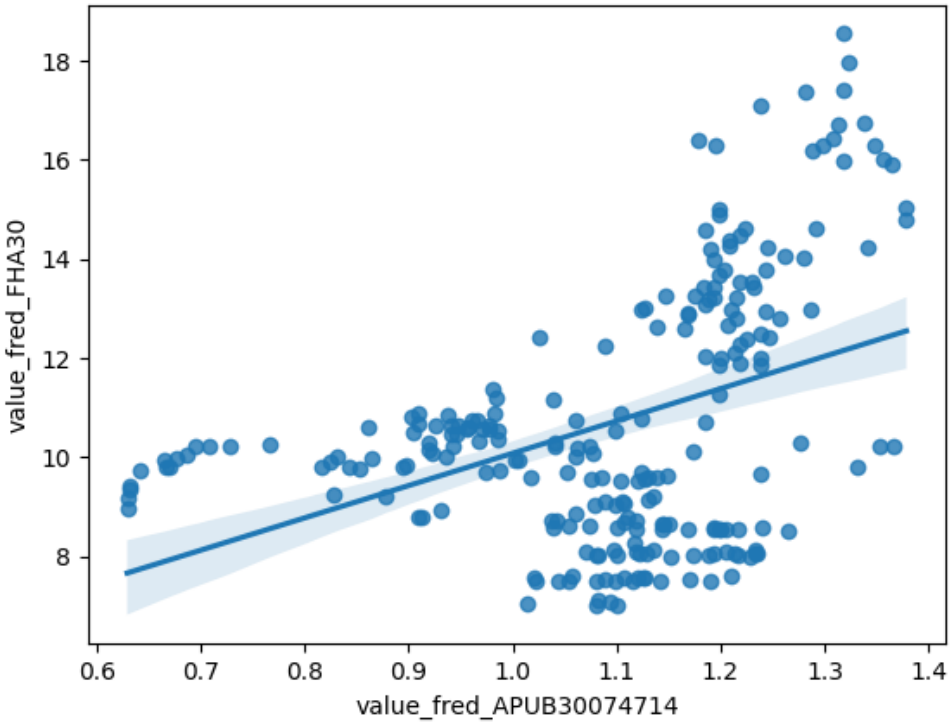


Figure 3: Raw Regression Plot for 2026-10-13

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_FHA30_detrended	**R-squared:**	0.626
Model:	OLS	**Adj. R-squared:**	0.624
Method:	Least Squares	**F-statistic:**	389.6
Date:	Tue, 13 Oct 2026	**Prob (F-statistic):**	1.23e-51
Time:	18:35:32	**Log-Likelihood:**	-352.60
No. Observations:	235	**AIC:**	709.2
Df Residuals:	233	**BIC:**	716.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	5.561e-13	0.071	7.82e-12	1.000	-0.140	0.140
value_fred_APUB30074714_detrended	8.7393	0.443	19.738	0.000	7.867	9.612

Omnibus:	5.577	**Durbin-Watson:**	0.209
Prob(Omnibus):	0.062	**Jarque-Bera (JB):**	5.236
Skew:	0.334	**Prob(JB):**	0.0729
Kurtosis:	3.299	**Cond. No.**	6.23

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

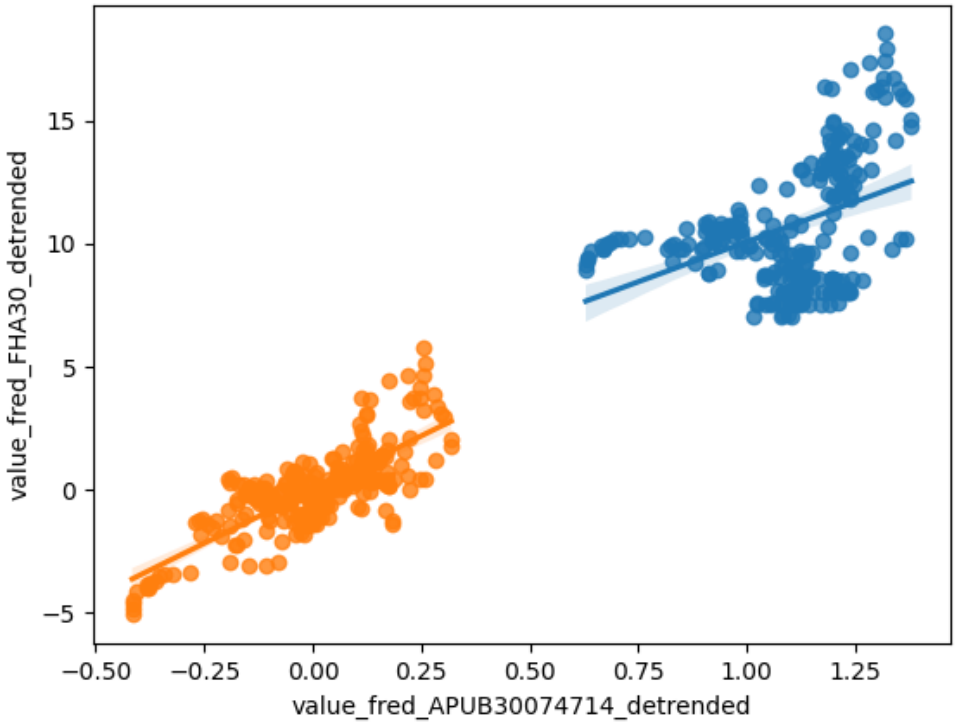


Figure 4: Detrended Regression Plot for 2026-10-13